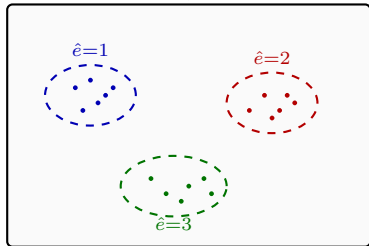


Emitter branch

Emitter embeddings $\{z_n^{\text{em}}\}_{n=1}^L \subset \mathbb{R}^p$

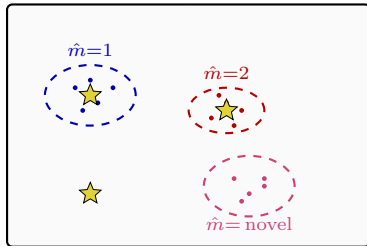


HDBSCAN per window

Train-local emitter IDs
 $\hat{e}_n \in \{1, \dots, \hat{K}_{\text{em}}\}$

Mode branch

Mode embeddings $\{z_n^{\text{md}}\}_{n=1}^L \subset \mathbb{R}^q$



HDBSCAN per window

Centroid matching to
catalogue prototypes

Mode IDs $\hat{m}_n \in \{1, \dots, M\} \cup \{\text{novel}\}$